

ROLL NO:250701573

EXPERIMENT NO: 2.DATA PREPROCESSING AND VISUALISATION.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('student_performance.csv')

print(df.head())

print(df.isnull().sum())

df['Marks'] = df['Marks'].fillna(df['Marks'].mean())
df['Attendance'] = df['Attendance'].fillna(df['Attendance'].mean())

print(df.describe())

summary = df.groupby('Department')[['Marks', 'Attendance']].mean()
print(summary)

plt.bar(summary.index, summary['Marks'])
plt.xlabel('Department')
plt.ylabel('Average Marks')
plt.title('Average Marks by Department')
plt.show()
```

```
plt.plot(df['Student_ID'], df['Marks'], marker='o')
plt.xlabel('Student ID')
plt.ylabel('Marks')
plt.title('Student Marks')
plt.show()
```

```
pivot = df.pivot_table(
    values='Marks',
    index='Department',
    columns='Subject',
    aggfunc='mean',
    fill_value=0
)
print(pivot)
```

```
correlation = df.corr(numeric_only=True)
print(correlation)
```

```
sns.heatmap(correlation, annot=True)
plt.title('Correlation Matrix')
plt.savefig('bar_grapg.png')
plt.show()
```

OUTPUT:

Student_ID	Name	Gender	Subject	Marks	Attendance	Department	
0	101	Arun	Male	Python	85	92	CSE
1	102	Priya	Female	Python	78	88	CSE

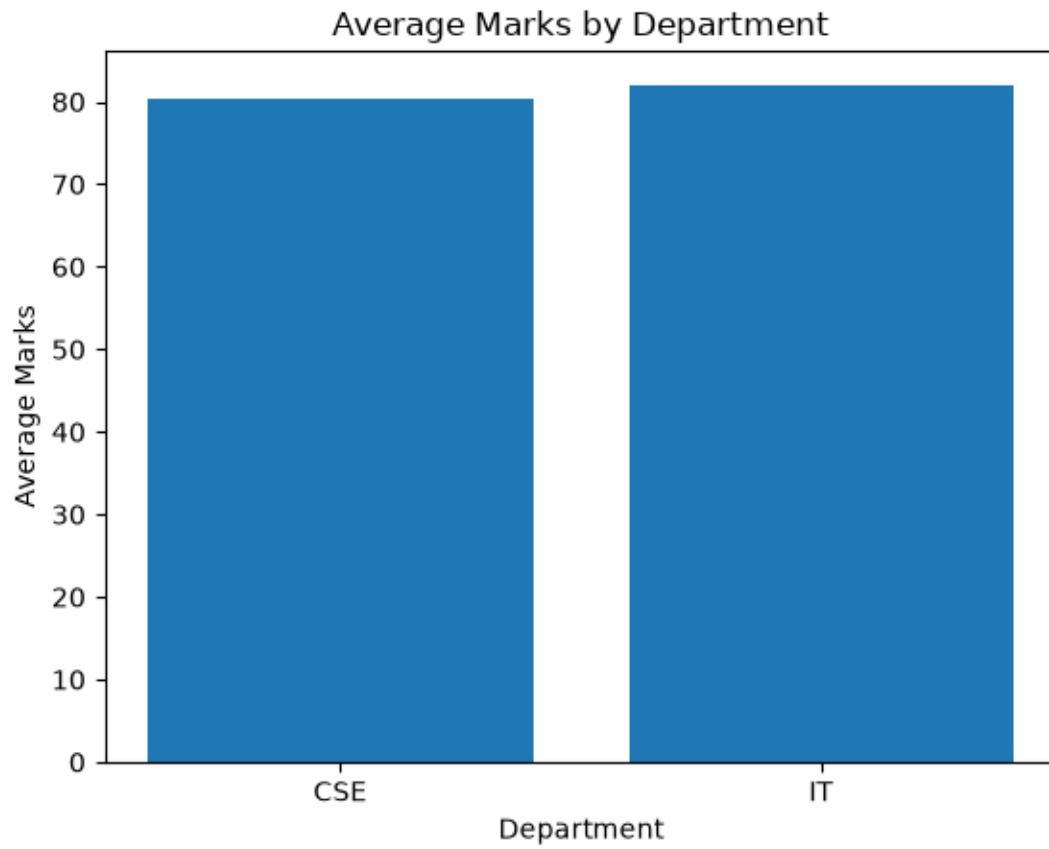
2	103	Kavin	Male	Java	91	95	IT
3	104	Divya	Female	Java	67	82	CSE
4	105	Suresh	Male	Python	73	79	IT

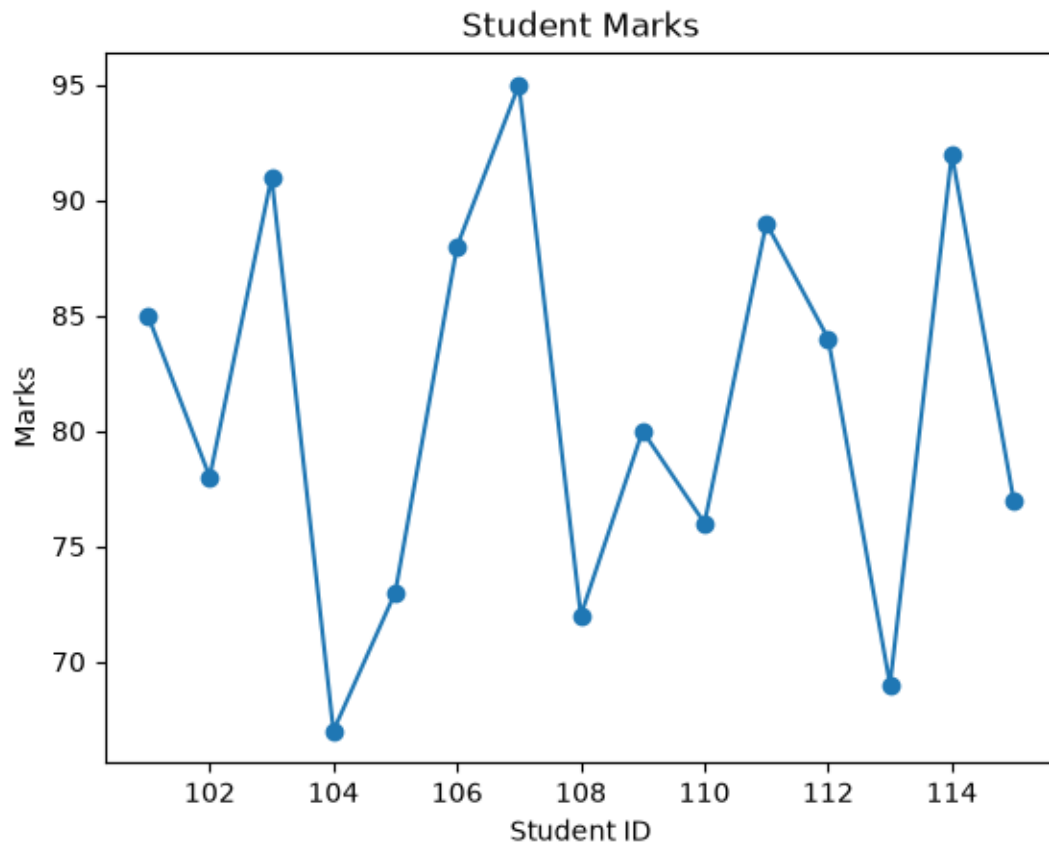
Student_ID 0
 Name 0
 Gender 0
 Subject 0
 Marks 0
 Attendance 0
 Department 0

dtype: int64

	Student_ID	Marks	Attendance
count	15.000000	15.000000	15.000000
mean	108.000000	81.066667	87.666667
std	4.472136	8.827446	5.814596
min	101.000000	67.000000	78.000000
25%	104.500000	74.500000	83.500000
50%	108.000000	80.000000	88.000000
75%	111.500000	88.500000	92.500000
max	115.000000	95.000000	96.000000

	Marks	Attendance
Department		
CSE	80.25	87.625000
IT	82.00	87.714286





Subject	Java	Python	
Department			
CSE	79.833333	81.5	
IT	91.000000	80.5	
	Student_ID	Marks	Attendance
Student_ID	1.000000	-0.009047	-0.131850
Marks	-0.009047	1.000000	0.930059
Attendance	-0.131850	0.930059	1.000000

